

Lab: Azure App Configuration

To Begin with the Lab

Summary

Azure App Configuration is a managed Azure service used to centrally store and manage application settings, feature flags, and configuration values. It enables applications to retrieve configuration dynamically without changing the application code or redeploying the application.

Understanding

Azure App Configuration provides a centralized place to store configuration settings for multiple applications and environments. Instead of hardcoding values such as database connection strings, API URLs, or feature flags inside the application, developers store them in Azure App Configuration. Applications can securely retrieve these settings at runtime, making configuration management easier, more secure, and more flexible. It also supports labels to manage different environments such as Development, Testing, and Production.

Example:

Suppose an application connects to different databases in Development and Production environments.

- **Development**
 - DatabaseServer = devserver.database.windows.net
 - FeatureX = Disabled
- **Production**
 - DatabaseServer = prodserver.database.windows.net
 - FeatureX = Enabled

Instead of modifying the application code, Azure App Configuration automatically provides the correct configuration based on the selected environment.

Lab Steps

Step 1: Open Azure Portal

- Sign in to the Azure Portal.
- Search for **App Configuration**.
- Open the App Configuration service.
- Click **Create**.

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Create a resource

Marketplace

Get Started

Explore solutions

Management

Private Marketplace

Private Offer Management

My Marketplace

Favorites

My solutions

Recently created

Private plans

Categories

Infrastructure Services (209)

IT & Management Tools (74)

app configuration

Pricing: All X Operating System: All X Publisher name: All X

☐ Azure benefit eligible only ⓘ ☐ Azure services only ☐ Free trial

Showing 1 to 20 of 327 results for 'app configuration'. [Clear search](#)



App Configuration

Microsoft

Azure Service

Create an app configuration to manage application settings in a centralized manner.

Create



App Service Environment v3

Microsoft

Azure Service

Deploy Azure App Services in your Virtual Network with greater scaling options

Create



Infoblox App for Microsoft Sentinel

Infoblox

Azure Application

Uplift your SIEM and SOAR Efficiency with Microsoft Sentinel and Infoblox

Price varies

Create



BriefCourt Web App

E Lex Ratio

Azure Application

Portal app for Brief Court Management System

Starts at Free

Create

Step 2: Create an App Configuration Store

- Select the **Azure Subscription**.
- Choose an existing **Resource Group** or create a new one.
- Enter a unique App Configuration Store name.
- Select the desired Region **East Asia**.
- Choose the Pricing Tier **Free**.
- Click **Review + Create**.

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Create a resource > Marketplace

Create App Configuration

these components can lead to hard-to-troubleshoot errors during an application deployment. Use App Configuration to store all the settings for your application and secure their accesses in one place. [Learn more](#)

Project Details

Subscription * Visual Studio Enterprise Subscription

Resource group * webapp

Create new

Instance Details

Location * East Asia

Resource name * ApplicationConfig420

Resource names are reserved for a period of time after deletion. [Learn more](#)

Pricing tier (View pricing details) * Free

Geo-replication is not supported for stores in the free pricing tier.

Geo-replication

Use the geo-replication feature to create replicas in other locations of your current configuration store for enhanced resiliency and availability.

Review + create < Previous Next: Access settings >

- Click **Create**.

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Create a resource > Marketplace

Create App Configuration

Validation passed

Basics Access settings Advanced Networking Encryption Tags Review + create

Basics

Subscription	Visual Studio Enterprise Subscription
Resource group	webapp
Resource name	ApplicationConfig420
Location	East Asia
Pricing tier	Free
Geo-replication	None

Access settings

Access keys	Disabled
Authentication mode	Pass-through

Advanced

Soft-delete	Disabled
Purge protection	Disabled
Days to retain deleted store	0 days
Days to retain revisions	7 days

Create < Previous Next >

Step 3: Open the App Configuration Store

- Wait until deployment completes.
- Click **Go to Resource**.

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home

Microsoft.AppConfiguration-20260716103305-8453 | Overview

Deployment

Search x « Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name : Microsoft.AppConfiguration-20260716103305-8453 Start time : 7/16/2026, 10:33:31 AM

Subscription : Visual Studio Enterprise Subscription Correlation ID : ede1b55a-a636-4f04-8d63-d150a395994f

Resource group : webapp

> Deployment details

Next steps

Go to resource

- Open the App Configuration Store dashboard.

Microsoft Azure

Home > Resource Manager | Resource groups > webapp

ApplicationConfig420 App Configuration

Search

Upgrade Delete Feedback

Overview

Activity log

Access control (IAM)

Tags

Quick start

Resource visualizer

Events

Configuration management

Configuration explorer

Configuration snapshots

Deliver configuration globally at hyperscale via [Azure Front Door](#). [Learn more](#)

Essentials

Resource group (move) : webapp

Status : Succeeded

Location : East Asia

Subscription (move) : Visual Studio Enterprise Subscription

Subscription ID : 294539c9-c109-4db8-b785-0043716c0b11

Endpoint : https://applicationconfig420.azureconfig.io

Pricing tier : Free (Click to upgrade)

Soft-delete : N/A

Purge protection : N/A

Geo-replication : N/A

Telemetry : Disabled

Tags (edit) : Add tags

Monitoring Capabilities Next steps Feedback

Step 4: Add Configuration Settings

- Select **Configuration Explorer**.
- Click **Create**.
- Choose the **Key-value**

Home > Resource Manager | Resource groups > webapp > ApplicationConfig420

ApplicationConfig420 | Configuration explorer

Search

Create Refresh Manage view Collapse Expand Values Advanced edit Delete Feedback

Overview

Activity log

Access control (IAM)

Tags

Quick start

Resource visualizer

Events

Configuration management

Configuration explorer

Configuration snapshots

Restore

Compare

Import/export

Feature management

Settings

Monitoring

Automation

Help

Deliver configuration globally at hyperscale via [Azure Front Door](#). [Learn more](#)

Labels: Select label Tags: Select tags Date: Select date

Default view

Loaded 0 key-values with 0 unique keys.

Key	Value	Label	Last modified	Content type
No results.				

- Enter a Key name.
- Enter the corresponding Value is **Welcome to the CloudXesu Orders**.
- Add a Label if required so, I have no required.
- Click on the **Apply**.

Home > Resource Manager | Resource groups > webapp > ApplicationConfig420

ApplicationConfig420 | Configuration explorer

Search

Create Refresh Manage view Collapse Expand Values Advanced edit Delete Feedback

Overview

Activity log

Access control (IAM)

Tags

Quick start

Resource visualizer

Events

Configuration management

Configuration explorer

Configuration snapshots

Restore

Compare

Import/export

Feature management

Settings

Monitoring

Automation

Help

Deliver configuration globally at hyperscale via [Azure Front Door](#). [Learn more](#)

Filter all key-values: Keys: Select key Labels: Select label Tags: Select tags Date: Select date

Search displayed key-values

Loaded 0 key-values with 0 unique keys.

Key	Value	Label
No results.		

Create

Create a new key-value.

Key * Order:BannerMessage

Value Welcome to the CloudXesu Orders

Label (No label)

Content type

Apply Cancel

Step 5: Add Package

- Install **dotnet-sdk**

- Open the own project folder in VS code
- Go to the Terminal
- Run this command

dotnet add package Microsoft.Extensions.Configuration.AzureAppConfiguration

```

PS C:\Users\rrath\Downloads\100.+Code\100. Code\05. Lab - Azure App Configuration - Configuration Settings\webapp> dotnet add package Microsoft.Extensions.Configuration.AzureAppConfiguration
info : X.509 certificate chain validation will use the default trust store selected by .NET for timestamping.
info : Adding PackageReference for package 'Microsoft.Extensions.Configuration.AzureAppConfiguration' into project 'C:\Users\rrath\Downloads\100.+Code\100. Code\05. Lab - Azure App Configuration - Configuration Settings\webapp\webapp.csproj'.
info : GET https://api.nuget.org/v3/registrations-gz-semester2/microsoft.extensions.configuration.azureappconfiguration/index.json
info : OK https://api.nuget.org/v3/registrations-gz-semester2/microsoft.extensions.configuration.azureappconfiguration/index.json 138ms
info : Restoring packages for C:\Users\rrath\Downloads\100.+Code\100. Code\05. Lab - Azure App Configuration - Configuration Settings\webapp\webapp.csproj...
info : GET https://api.nuget.org/v3/vulnerabilities/index.json 95ms
info : OK https://api.nuget.org/v3/vulnerabilities/index.json 95ms
info : GET https://api.nuget.org/v3-vulnerabilities/2026.07.16.05.27.22/vulnerability.base.json
info : GET https://api.nuget.org/v3-vulnerabilities/2026.07.16.05.27.22/vulnerability.update.json
info : OK https://api.nuget.org/v3-vulnerabilities/2026.07.16.05.27.22/vulnerability.base.json 310ms
info : OK https://api.nuget.org/v3-vulnerabilities/2026.07.16.05.27.22/vulnerability.update.json 416ms
info : Package 'Microsoft.Extensions.Configuration.AzureAppConfiguration' is compatible with all the specified frameworks in project 'C:\Users\rrath\Downloads\100.+Code\100. Code\05. Lab - Azure App Configuration - Configuration Settings\webapp\webapp.csproj'.
info : PackageReference for package 'Microsoft.Extensions.Configuration.AzureAppConfiguration' version '8.5.0' updated in file 'C:\Users\rrath\Downloads\100.+Code\100. Code\05. Lab - Azure App Configuration - Configuration Settings\webapp\webapp.csproj'.
info : Assets file has not changed. Skipping assets file writing. Path: C:\Users\rrath\Downloads\100.+Code\100. Code\05. Lab - Azure App Configuration - Configuration Settings\webapp\obj\project.assets.json
log : Restored C:\Users\rrath\Downloads\100.+Code\100. Code\05. Lab - Azure App Configuration - Configuration Settings\webapp\webapp.csproj (in 2.05 sec).
PS C:\Users\rrath\Downloads\100.+Code\100. Code\05. Lab - Azure App Configuration - Configuration Settings\webapp>

```

- Run the next command

dotnet add package Azure.Data.AppConfiguration

```

PS C:\Users\rrath\Downloads\100.+Code\100. Code\05. Lab - Azure App Configuration - Configuration Settings\webapp> dotnet add package Azure.Data.AppConfiguration
info : X.509 certificate chain validation will use the default trust store selected by .NET for code signing.
info : X.509 certificate chain validation will use the default trust store selected by .NET for timestamping.
info : Adding PackageReference for package 'Azure.Data.AppConfiguration' into project 'C:\Users\rrath\Downloads\100.+Code\100. Code\05. Lab - Azure App Configuration - Configuration Settings\webapp\webapp.csproj'.
info : GET https://api.nuget.org/v3/registrations-gz-semester2/azure.data.appconfiguration/index.json
info : OK https://api.nuget.org/v3/registrations-gz-semester2/azure.data.appconfiguration/index.json 590ms
info : Restoring packages for C:\Users\rrath\Downloads\100.+Code\100. Code\05. Lab - Azure App Configuration - Configuration Settings\webapp\webapp.csproj...
info : GET https://api.nuget.org/v3-flatcontainer/azure.data.appconfiguration/index.json
info : OK https://api.nuget.org/v3-flatcontainer/azure.data.appconfiguration/index.json 583ms
info : GET https://api.nuget.org/v3-flatcontainer/azure.data.appconfiguration/1.10.0/azure.data.appconfiguration.1.10.0.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/azure.data.appconfiguration/1.10.0/azure.data.appconfiguration.1.10.0.nupkg 153ms
info : GET https://api.nuget.org/v3-flatcontainer/azure.core/index.json
info : OK https://api.nuget.org/v3-flatcontainer/azure.core/index.json 143ms
info : GET https://api.nuget.org/v3-flatcontainer/azure.core/1.59.0/azure.core.1.59.0.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/azure.core/1.59.0/azure.core.1.59.0.nupkg 693ms
info : GET https://api.nuget.org/v3-flatcontainer/microsoft.bcl.asyncinterfaces/index.json
info : GET https://api.nuget.org/v3-flatcontainer/microsoft.identity.client.extensions.msal/index.json
info : GET https://api.nuget.org/v3-flatcontainer/microsoft.identity.client/index.json

```

- Run the next command

dotnet build

```

PS C:\Users\rrath\Downloads\100.+Code\100. Code\05. Lab - Azure App Configuration - Configuration Settings\webapp> dotnet build
Restore complete (0.4s)
webapp net10.0 succeeded (1.3s) -> bin\Debug\net10.0\webapp.dll
Build succeeded in 2.2s
PS C:\Users\rrath\Downloads\100.+Code\100. Code\05. Lab - Azure App Configuration - Configuration Settings\webapp>

```

Step 6: Add connection string

- Go to the just created **App Configuration**
- Navigate to **App Setting**
- Click on the **Values**
- Copy the **connection string**

Search Values Feedback

Overview

Activity log

Access control (IAM)

Tags

Quick start

Resource visualizer

Events

Configuration management

Configuration explorer

Configuration snapshots

Restore

Compare

Import/export

Feature management

Settings

Access settings

Pricing tier

Access keys

Read-write keys Read-only keys

Endpoint

https://applicationconfig420.azureconfig.io

Primary key

Regenerate

Id (credential)

Secret

Connection string

Secondary key

Regenerate

Id (credential)

Copied

- Go to the own Webapp
- Navigate to Environment Variables

Microsoft Azure Search resources, services, and docs (G+/)

Home

web-dev-esu-12 Web App

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Microsoft Defender for Cloud

Events (preview)

AI (preview)

Resource visualizer

Deployment

Deployment slots

Deployment Center

Settings

Environment variables

Configuration

Essentials

Resource group (move) : webapp

Status : Running

Location (move) : East US

Subscription (move) : Visual Studio Enterprise Subscription

Subscription ID : 294539c9-c109-4db8-b785-0043716c0b11

Tags (edit) : Add tags

Default domain : web-dev-esu-12-brgbsdbdbdgmhnm.eastus-01.azurewebsites.net

App Service Plan : ASP-webapp-930f (\$1: 1)

Operating System : Windows

Health Check : Not Configured

Properties

Monitoring Logs Capabilities Notifications (0) Recommendations

Web app

Name : web-dev-esu-12

Publishing model : Code

Runtime Stack : Dotnet - v10.0

Deployment Center

Deployment logs : View logs

Last deployment : Successful on Thursday, July 16, 10:14:25 AM Refresh

Deployment provider : None

Domains

Default domain : web-dev-esu-12-brgbsdbdbdgmhnm.eastus-

Application Insights

- Click on the Connection string
- Click Add

Microsoft Azure Search resources, services, and docs (G+/)

Home > web-dev-esu-12

web-dev-esu-12 | Environment variables

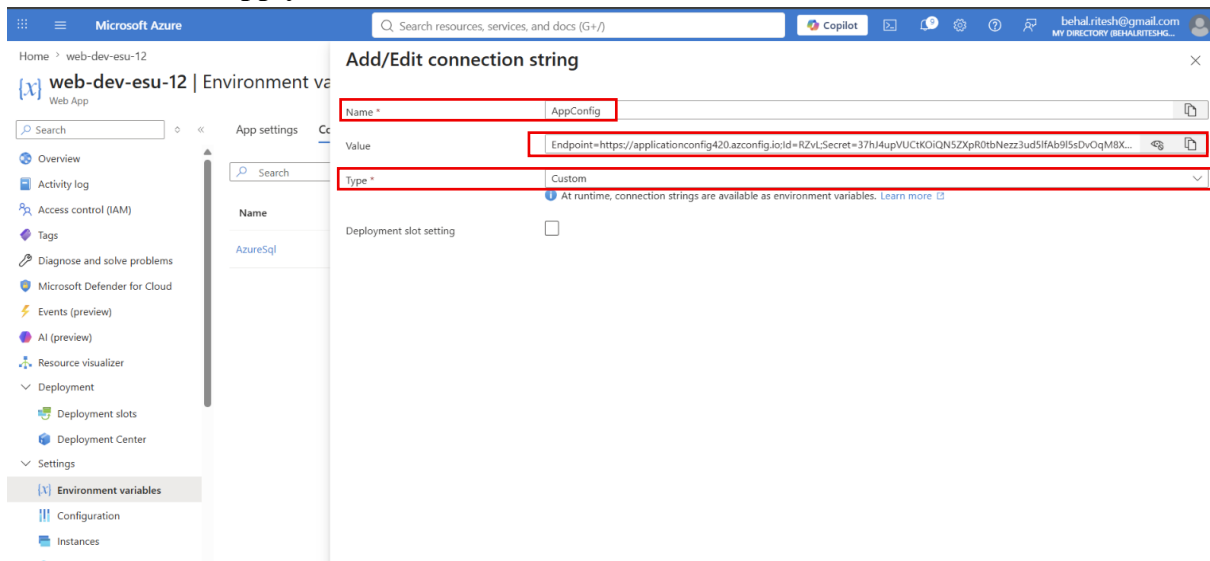
App settings Connection strings

Search Add Refresh Show values Advanced edit Pull reference values

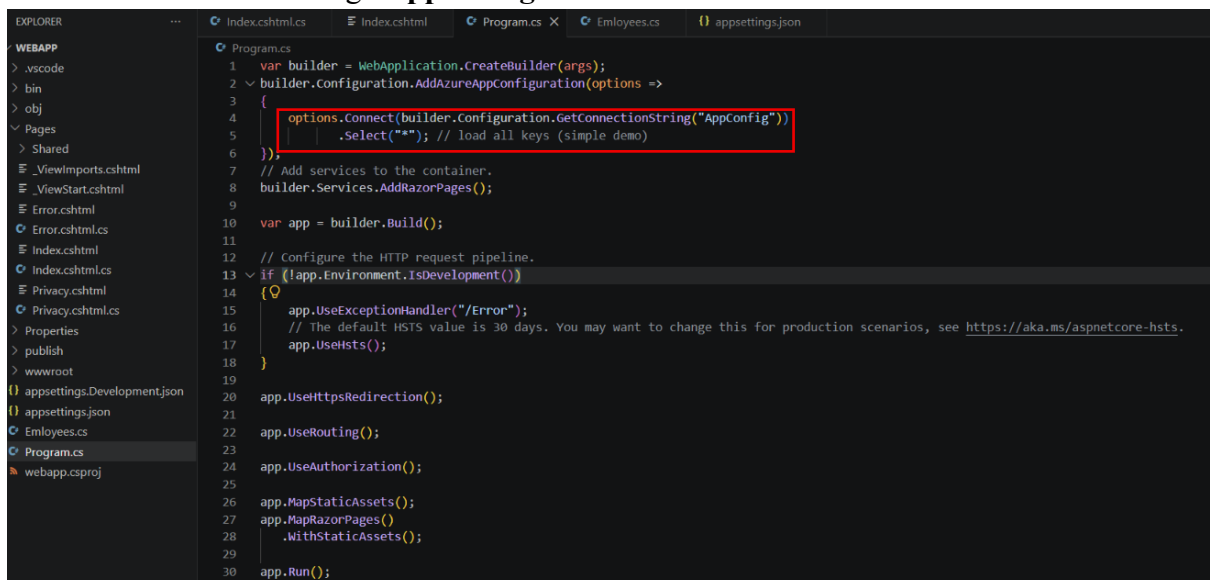
Name	Value	Deployment slot setting	Type	Source	Delete
AzureSql	Show value	✓	SQLAzure	App Service	Delete

- Enter the Name AppConfig

- Paste the **Connection string**
- Choose type **custom**
- Click on the **Apply**



- Go to the **VS code**
- Some changes in **Program.cs** file
- Add a connection string **“AppConfig”**



- Some Changes in **Index.cshtml.cs** file
- Add **BannerMessage**

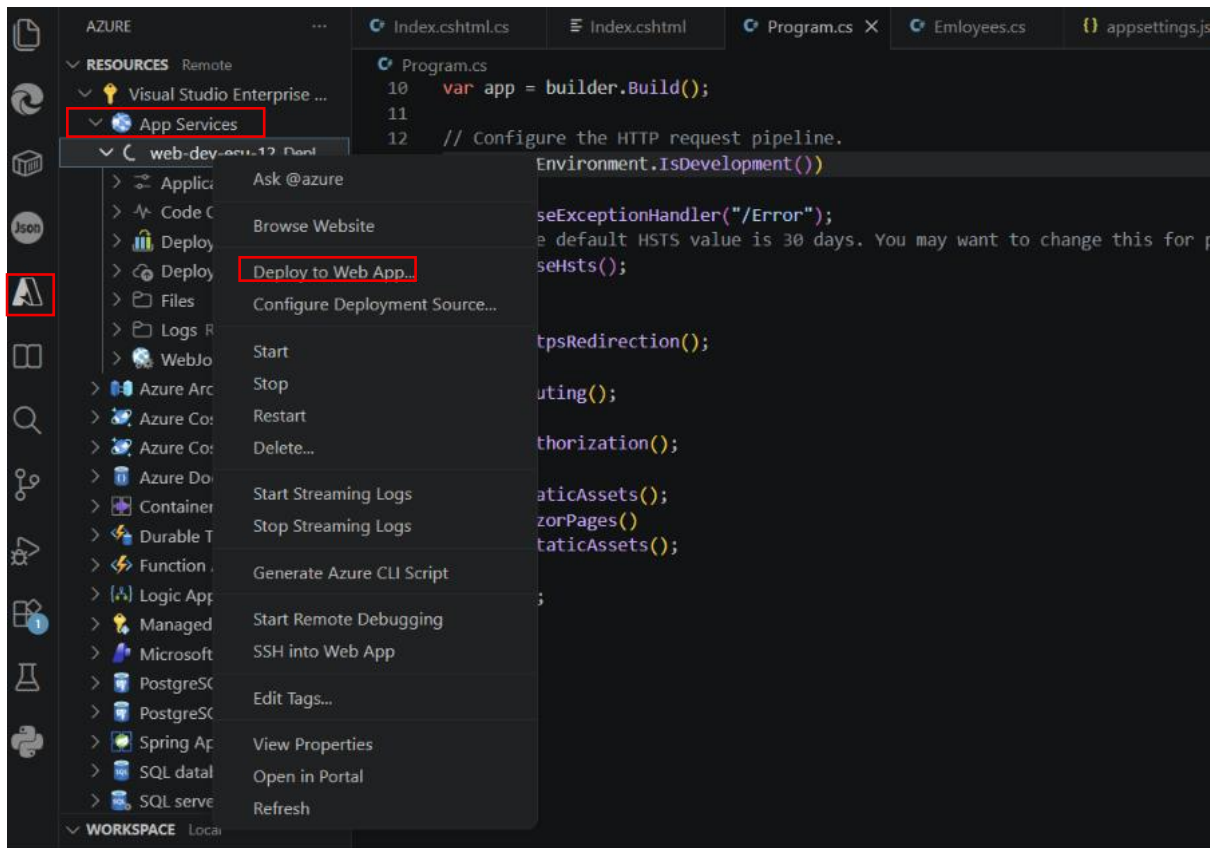
```
5 namespace webapp.Pages.Orders;
7 public class IndexModel : PageModel
{
12
13     public IndexModel(IConfiguration config)
14     {
15         _config = config;
16     }
17
18     3 references
19     public List<Employee> Employees { get; private set; } = new();
20
21     0 references
22     public async Task OnGetAsync()
23     {
24         string? connString = _config.GetConnectionString("Azuresql");
25         BannerMessage = _config["Orders:BannerMessage"];
26
27         using var conn = new SqlConnection(connString);
28         await conn.OpenAsync();
29
30         string sql = @"
31         SELECT
            Id,
            Name,
```

- Some add code in **Index.cshtml** file

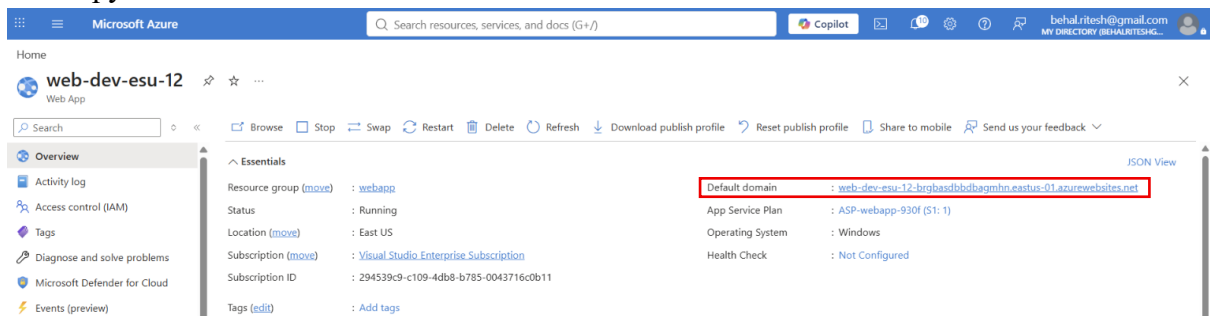
```
@if (!string.IsNullOrEmpty (Model.BannerMessage))
{
<p>@Model.BannerMessage</p>
}
```

```
1 @page
2 @model webapp.Pages.Orders.IndexModel
3
4 <h2>Employee List</h2>
5 <h2>Course Orders</h2>
6 @if (!string.IsNullOrEmpty (Model.BannerMessage))
7 {
8     <p>@Model.BannerMessage</p>
9 }
10 @if (Model.Employees.Count == 0)
11 {
12     <p>No employees found.</p>
13 }
```

- Go to the Azure in **VS code**
- Navigate to **App Service**
- **Right-click** on the WebApp
- Click on the **Deploy**
- Wait until a few minutes



- Go to the **own Webapp** in Azure
- Copy the **Default Domain**



- Paste the **Browser**
- Added the **Banner message** in webpage

